

Take Home Instructions

Premise:

An analyst tells you that they need data from an NHL website. They explain that every week they manually copy/paste data from the website into Excel and would like a more automated process. They haven't fully explained the details of how or what they need from that data and will be unavailable for clarification. After being given this task, you start exploring existing storage and you find a file which the NHL automatically generates every night, even better it is a .csv!

You know that analysts frequently ask which players have the most goals as well as who has the most penalties per minute. You can recall a discussion about which team had the most goals and shots in the league and how they ranked against other teams in the league. As you think back about other questions this data could answer, you recall other conversations about which players changed teams the most this season and the active rosters of a given team.

Assignment:

Using Docker, Python, and a database of your choosing, write an API in FastAPI with routes and endpoints that **help analysts answer questions about the data**. The database and API should be run via Docker Compose.

- Create a simple upload or insert process into the database with the given datafile.
- Conduct any data cleaning or modeling that you see fit.
- Once the service is up and running, consider how you may add robustness to the system. Focus on showcasing your strengths.

Submission:

Upload your project folder to the Box link that was provided to you. The submitted project must be reproducible, i.e. another engineer should be able to run `docker compose up` and have the full system working. Your submission must include:

1. All source code, Dockerfiles, and configuration files needed to stand up the project
2. README.md containing:
 - Architecture overview and database schema design
 - Documented examples of querying each endpoint with sample responses
 - Any struggles you came across
 - Estimated time spent on the project
3. AI Usage Disclosure: Use of AI coding tools is welcome and common practice on our team. In a dedicated section of your README:
 - Describe which AI tools you used and for what tasks
 - Identify areas where you accepted, modified, or rejected AI suggestions and why